

— Matched Dr.GRPO — E58/E59 verified-first + global split mass/balance replay

Current canonical experiment — E58 + E59 ModeBench three-domain sentinel + matched MathIR extension — 0.5B



Dr.GRPO and preregistered evaluation. Every x-axis expands with the latest treatment checkpoint; panels in a row share that live frontier, while metric ranges tighten to the values observed so far. E58 + E59 uses K=8 exact-answer metrics with a 50-pass sentinel budget, but live-chart ingestion remains uncapped and will display later checkpoints. Gold-support-normalized coverage is deliberately omitted from this decision view and never enters the controller or terminal gate. Cancelled prompt is not MATH-500. Bank admission is validator-bound in every treatment: Countdown uses the executed exact AST, graph coloring the constraint-checked vector, and Python factors the externally executed return vector, while MathIR uses the exact rational equation-state trajectory from the submitted and executed finite action sequence. Its held-out action-menu evaluation is not MATH-500. The model cannot supply its own canonical key. The first three rows retain the checkpointed prompt-hash round-robin order. Verified surprisal controls common mass, verified-bank entropy controls relative known-mode balance, and success-only open-set pressure explores newly verified outcomes. All three coefficients use only their own 64-observation warmup/EMA state and remain unbounded. The one-bank schedule is a fixed compute measure, not a desired entropy or mode count; training never reads gold support or evaluation. The first three rows retain the